

Phyloseq Visualizations in Galaxy

BIOL 4810

2026-04-16

Goal: Use **phyloseq in Galaxy** to visualize your microbial community data with **alpha diversity**, **beta diversity**, and a **taxonomic bar plot**.

Submission: Fill in answers, upload screenshots where prompted, then click **Prepare submission preview** and **Download as PDF (Print)**.

0) Your dataset

0A. Sample set / project name:

Bird microbiome

0B. Grouping variable used for comparisons:

phylum

0C. Other metadata variable(s) available for plotting or faceting:

infection status, sample source

phyloseq workflow focus

For this module, you will generate and interpret:

Alpha diversity → Beta diversity (NMDS using Bray-Curtis) → Taxonomic bar plot

Reminder: These visualizations help you describe diversity within samples, differences among samples, and which taxa are driving those patterns.

Part 1 — Alpha diversity

1A) Alpha diversity

In your own words, what is alpha diversity?

Alpha diversity measures and describes how diverse a particular community is.

1B) Observed richness

What does Observed measure?

Observed species richness simply measures the total number of species.

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1C) Shannon diversity

What does Shannon measure?

Shannon (alpha) diversity accounts for richness and evenness by comparing species abundance.

//

1D) Simpson diversity

What does Simpson measure?

Simpson (alpha) diversity accounts for richness and evenness with less sensitivity to rare species, essentially indicating the probability of picking the same species twice.

//

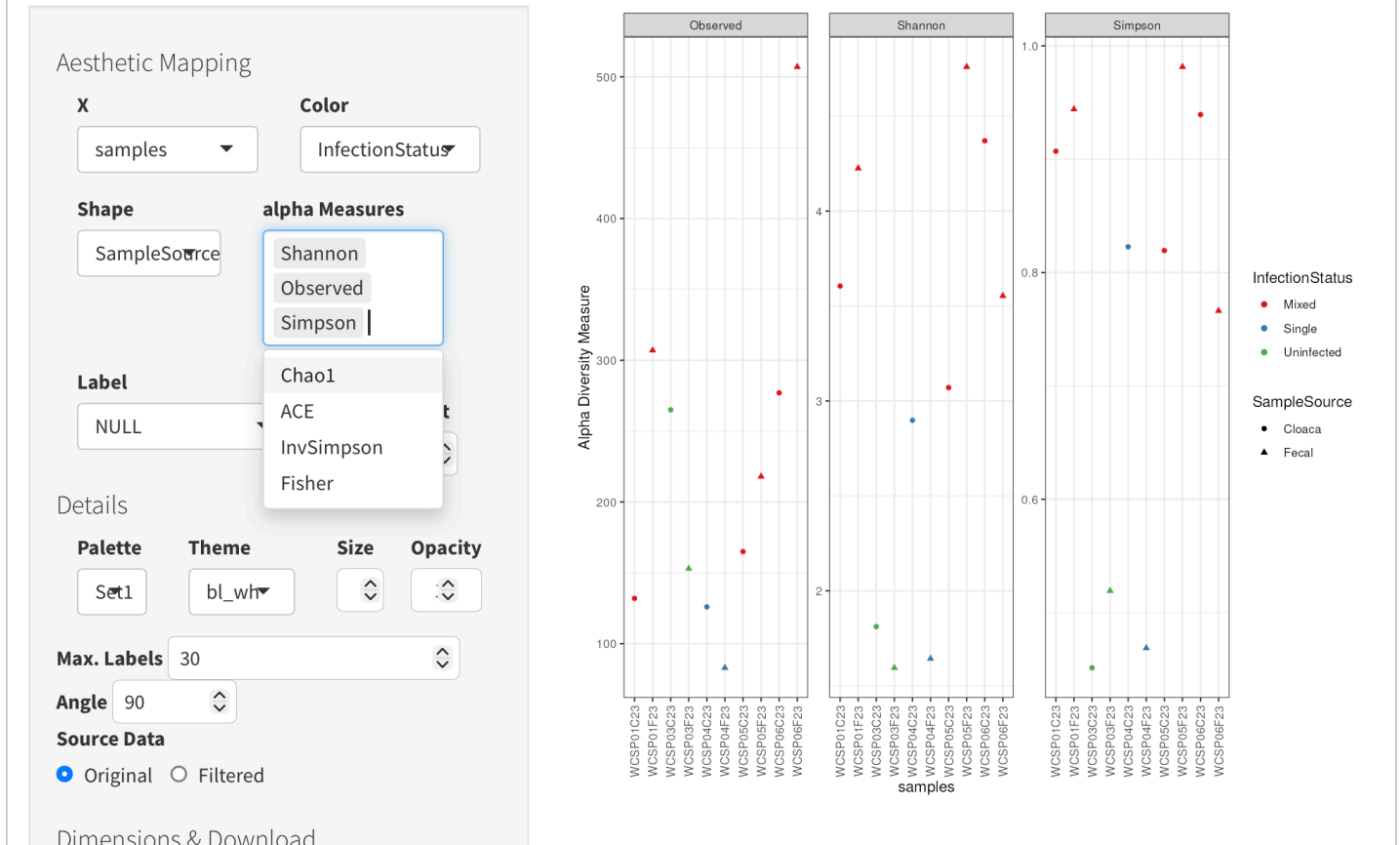
Upload screenshot: alpha diversity plot

Choose File



AlphaDiver...creenshot

Alpha Diversity Estimates



Grouping variable shown on the alpha diversity plot:

sample name

Which sample or group appears to have the greatest alpha diversity?

Sample WCSP06F23 appeared to have the greatest (observed) alpha diversity.

Did Observed, Shannon, and Simpson show the same pattern?

Unlike what's seen in the observed alpha diversity plot, sample WCSP06C23 appears to have the greatest alpha diversity in the Shannon and Simpson plots.

Part 2 — Beta diversity (NMDS with Bray-Curtis)

What beta diversity means:

Beta diversity describes differences in community composition **between samples**.

In your own words, what is beta diversity?

Beta diversity is a comparison of one community's composition to that of another.

2A) Bray-Curtis dissimilarity

What does Bray-Curtis measure?

individuals (represented by datapoints) with similar microbiomes (respective datapoints will be closer) or dissimilar (respective datapoints will be farther).

2B) NMDS

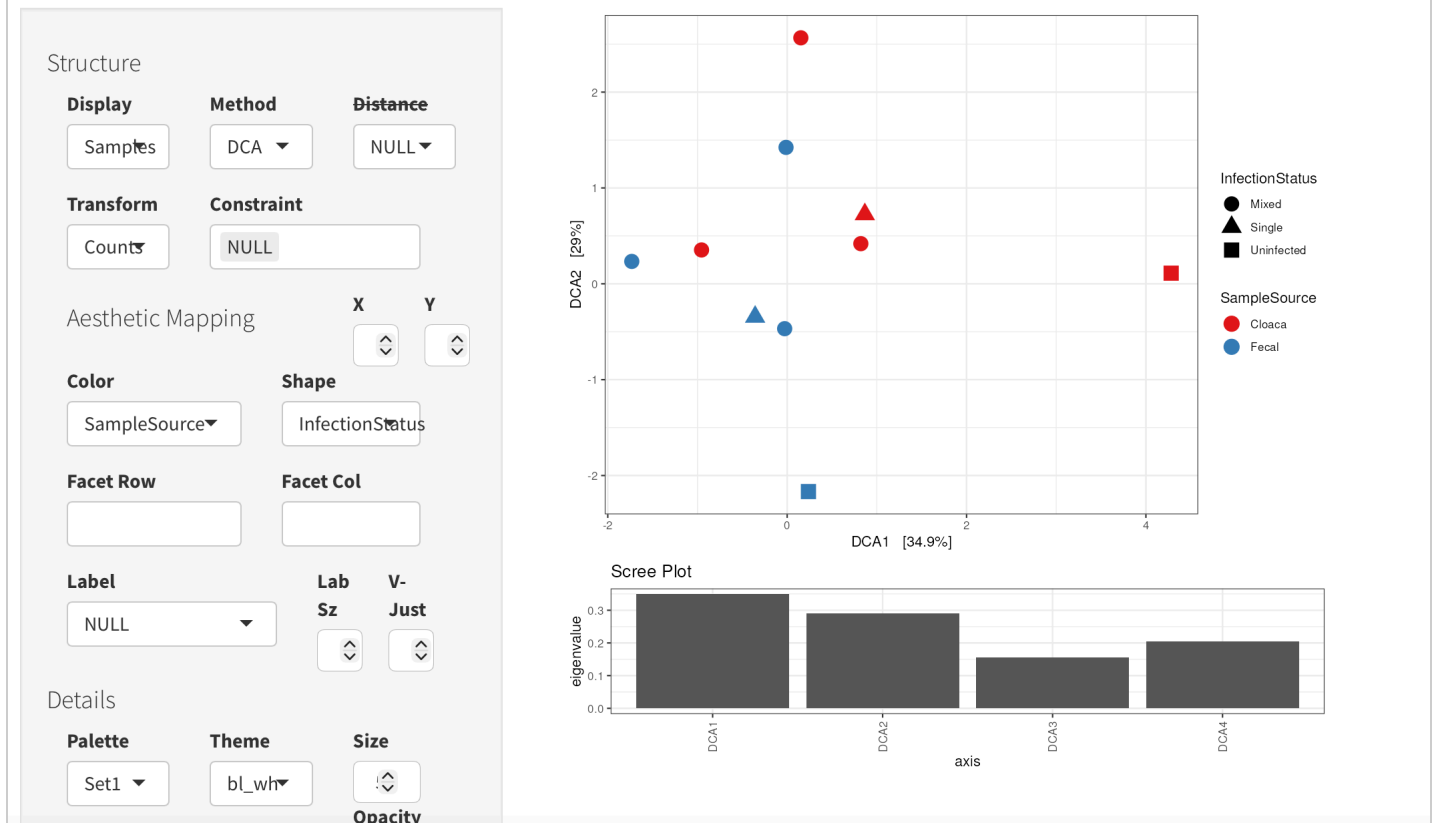
What does NMDS show?

The NMDS plot shows approximate similarity by sample source in this case, and more generally represents dissimilarity by datapoint proximity.

Upload screenshot: NMDS Bray-Curtis plot

Choose File  NMDSbraycurtis

Ordination Plot



Color variable used on the NMDS plot:

sample source

Shape variable used on the NMDS plot (if any):

infection status

Which samples or groups cluster together?

Sample source groups seem to cluster together more closely than infection status.

Are any groups clearly separated from one another? What might that suggest?

Separation between points based off infection status may suggest that such samples have different community composition.

If points overlap strongly, what does that mean?

Overlapping points indicate that samples are very similar in composition-- the communities possessed by those samples have similar taxa and relative abundances.

Part 3 — Taxonomic bar plot**What a taxonomic bar plot shows:**

A bar plot summarizes the composition of samples by showing the relative abundance of taxa at a selected taxonomic level.

What taxonomic level did you use?

phylum

How did you facet the plot?

faceted rows by infection status

Why was this taxonomic level useful?

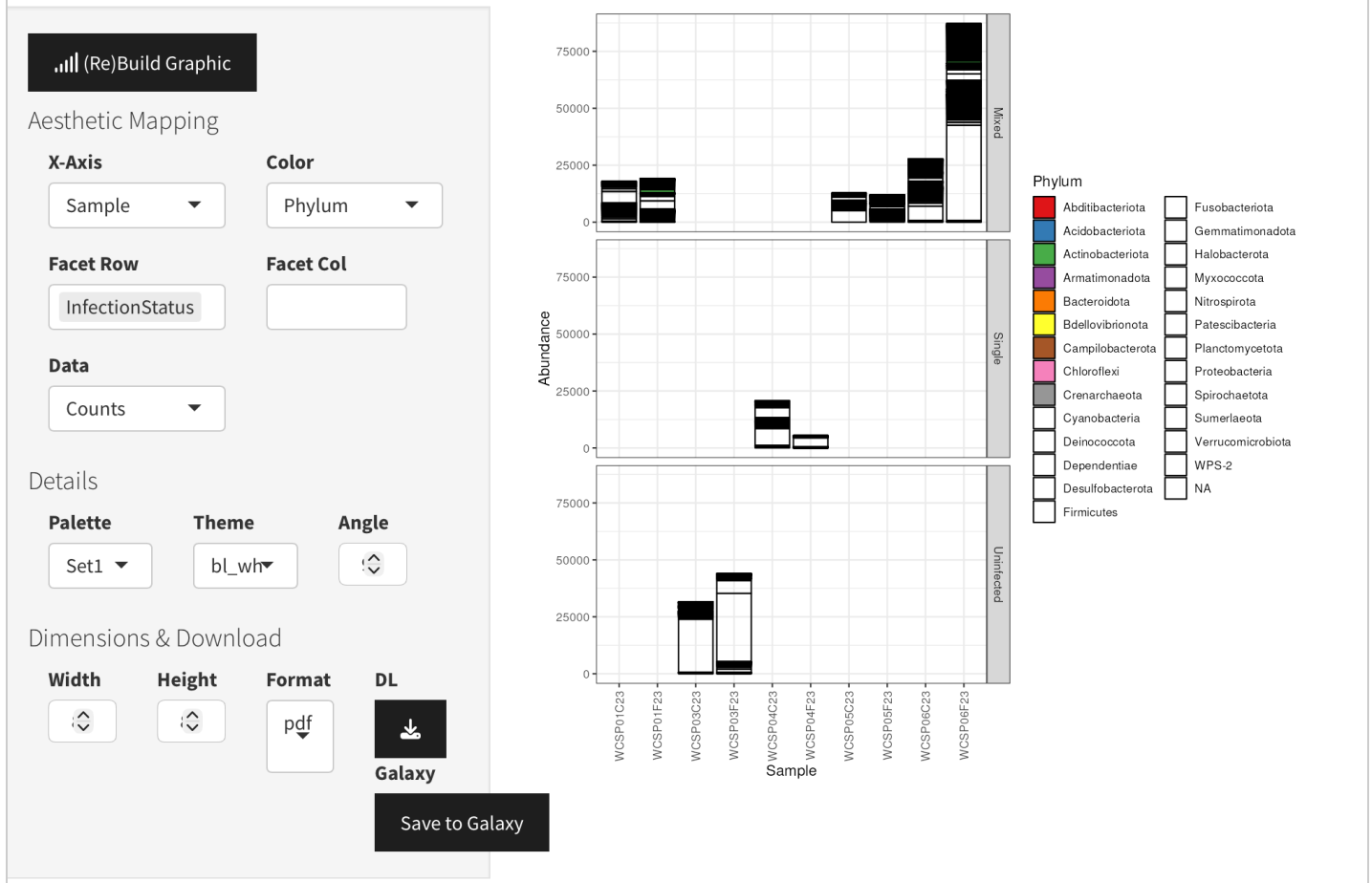
Using phylum as the taxonomic level here provides an overview of community composition without being too specific, so patterns may be more visible during interpretation.

Why was your faceting choice useful?

Faceting by infection status allows us to target/interpret differences in community composition based off infection status.

Upload screenshot: taxonomic bar plotChoose File  Taxonomic...screenshot

Flexible Bar Plot

**Which taxa appear to be most abundant?**

Having trouble interpreting-- will ask in office hours.

Did taxonomic composition differ among groups?

Abundance and taxonomic composition did differ among groups!

Part 4 — Overall interpretation

In 3–5 sentences, summarize what your alpha diversity, beta diversity, and bar plot together suggest about this dataset.

Bray-Curtis NMDS plot reveals that samples collected from the same source roughly share similar microbiomes. The taxonomic bar plot shows that the microbiomes of different samples vary on the basis of infection status-- mixed infection status meant similar phylum grouping and abundance. Alpha diversity also revealed that the highest species richness and evenness tended to correspond with samples of mixed infection status.

Export (PDF)

1. Click **Prepare submission preview** (generates a clean summary).
2. Click **Download as PDF (Print)** and save.

Submission preview

Dataset

Project/sample set: Bird microbiome

Grouping variable: phylum

Other metadata variables: infection status, sample source

Alpha diversity

Definition: Alpha diversity measures and describes how diverse a particular community is.

Observed: Observed species richness simply measures the total number of species.

Shannon: Shannon (alpha) diversity accounts for richness and evenness by comparing species abundance.

Simpson: Simpson (alpha) diversity accounts for richness and evenness with less sensitivity to rare species, essentially indicating the probability of picking the same species twice.

Grouping variable: sample name

Highest alpha diversity: Sample WCSP06F23 appeared to have the greatest (observed) alpha diversity.

Did the metrics show the same pattern? Unlike what's seen in the observed alpha diversity plot, sample WCSP06C23 appears to have the greatest alpha diversity in the Shannon and Simpson plots.

Beta diversity (NMDS Bray-Curtis)

Definition: Beta diversity is a comparison of one community's composition to that of another.

Bray-Curtis: Bray-Curtis (beta diversity) measures dissimilarity and is represented by ordination techniques to visualize individuals (represented by datapoints) with similar microbiomes (respective datapoints will be closer) or dissimilar (respective datapoints will be farther).

NMDS: The NMDS plot shows approximate similarity by sample source in this case, and more generally represents dissimilarity by datapoint proximity.

Color variable: sample source

Shape variable: infection status

Which groups cluster together? Sample source groups seem to cluster together more closely than

infection status.

Are any groups separated? Separation between points based off infection status may suggest that such samples have different community composition.

If points overlap strongly: Overlapping points indicate that samples are very similar in composition-- the communities possessed by those samples have similar taxa and relative abundances.

Taxonomic bar plot

Taxonomic level: phylum

Faceted by: faceted rows by infection status

Why this taxonomic level? Using phylum as the taxonomic level here provides an overview of community composition without being too specific, so patterns may be more visible during interpretation.

Why this faceting choice? Faceting by infection status allows us to target/interpret differences in community composition based off infection status.

Most abundant taxa: Having trouble interpreting-- will ask in office hours.

Did composition differ among groups? Abundance and taxonomic composition did differ among groups!

Overall interpretation

Bray-Curtis NMDS plot reveals that samples collected from the same source roughly share similar microbiomes. The taxonomic bar plot shows that the microbiomes of different samples vary on the basis of infection status-- mixed infection status meant similar phylum grouping and abundance. Alpha diversity also revealed that the highest species richness and evenness tended to correspond with samples of mixed infection status.